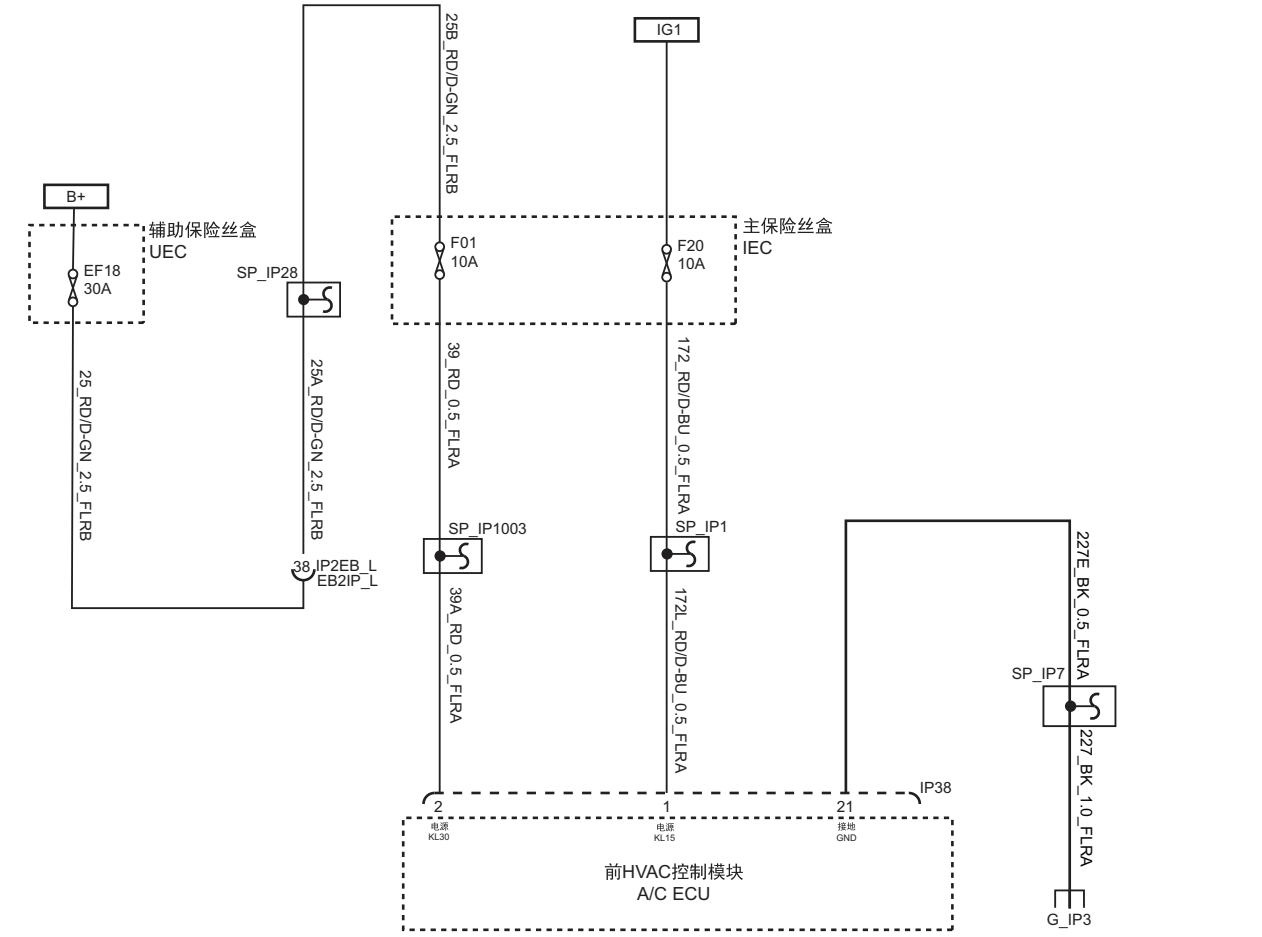
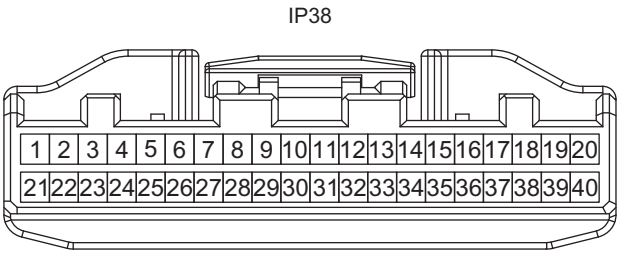
B1608-1C、 B1609-1C

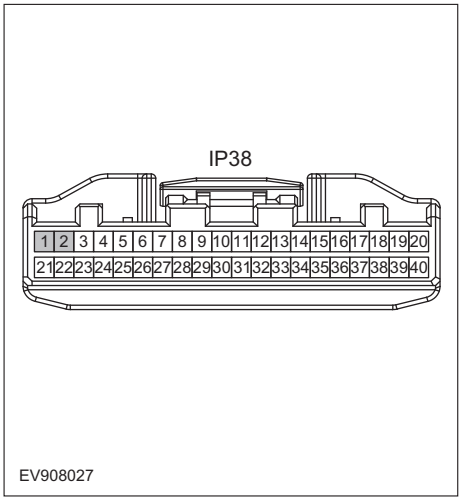
Description of fault diagnosis codes
B1608-1C：Voltage High
B1609-1C：Voltage Low
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • Battery • Air conditioning control module • Charging system • Insurance fuse
Conditions for fault code operation
The ignition switch is in ON position.
Conditions for fault code setting
B1608-1C
B1609-1C
To set the effect of a fault code condition
Description of circuit diagram
The air conditioning control module will monitor whether all sensors and actuators are within normal range all the time. Meantime, it will also detect virtual connection of joints, short circuit, open circuit and the rationality between other signals. When fault occurs, the air conditioning control module will save the fault code corresponding to that fault and enable safety mode.

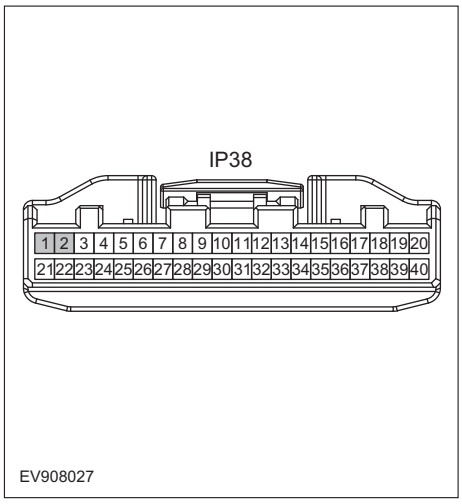
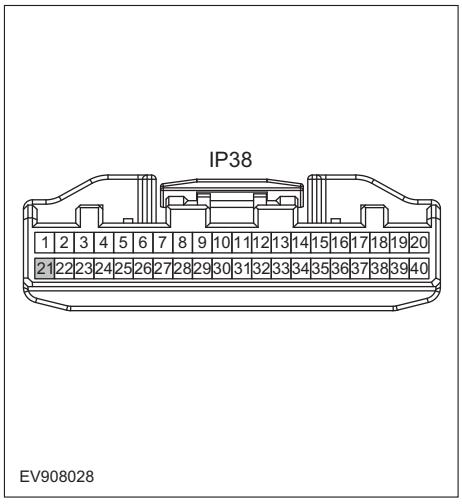
Circuit diagram





EV908026

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check EF18,F01,F20 fuses for damages. 2 Check battery capacity. 3 Check the harness plug of air conditioning control module for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON position. 3 Diagnose the air condition control system with diagnostic apparatus. 4 If there is any fault code except for B1608-1C,B1609-1C?
		YES → Refer to:DTC Summary List (HVAC).
NO ↓		
Step 3	Check the battery of air conditioning control module.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP38 of air conditioning module. 3 Measure the voltage between the air conditioning control module harness terminal and ground. Measure circuit: voltage between terminal 1 on plug IP38 and ground. Measure circuit: voltage between terminal 2 on plug IP38 and ground. **Standard value:10-14V** 4 Check whether the result is normal or not?
		NO → Please repair or replace the faulted parts.
YES ↓		

Step 4	Check the power voltage when the air conditioning control module is started or operating
	1 Turn the ignition switch to ON. 2 Measure the voltage between the air conditioning control module harness terminal and ground. Measure circuit: voltage between terminal 1 on plug IP38 and ground. Measure circuit: voltage between terminal 2 on plug IP38 and ground. **Standard value:10-14V** 3 Check whether the result is normal or not?
	NO → Please repair or replace the faulted parts.
YES ↓	
Step 5	Test whether the grounding circuit of air conditioning control module is open.
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug IP38 of air conditioning module. 3 Measure the resistance between harness plug of air conditioning control module and ground. Measuring circuit: resistance between terminal 21 on plug IP38 and ground. **Standard value: < 1Ω** 4 Test whether the resistance is less than 1Ω?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 6	Check if air conditioning module is normal.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug IP38 of air conditioning controller. 3 Check and repair; • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the air conditioning control module plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the air conditioning control module.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	